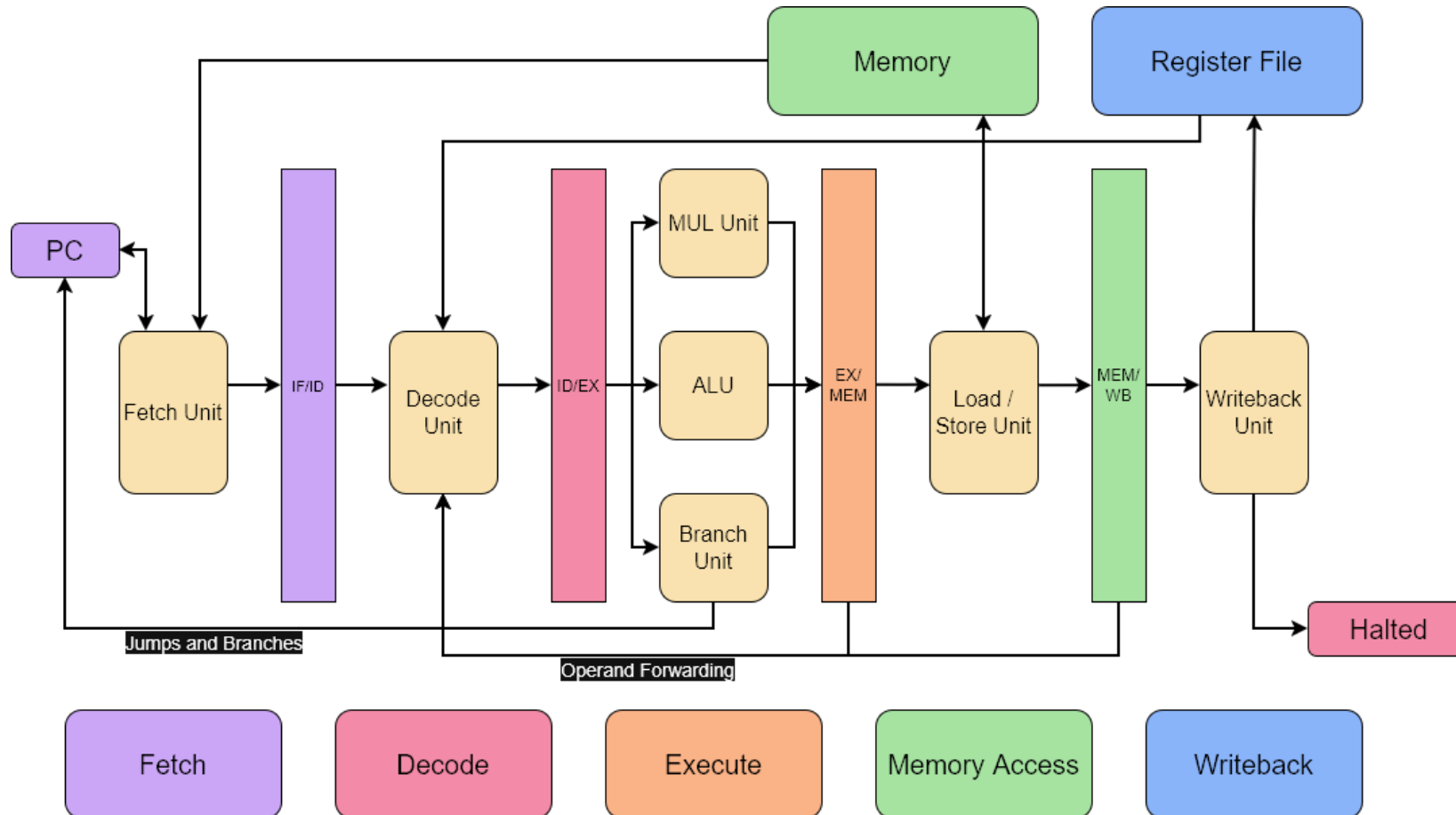


RISC-V Pipelined Scalar Simulator

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Microarchitecture



Features

- RV32im – implemented base unprivileged ISA with multiply extension
 - Omitted ‘FENCE’ instructions, ECALL and EBREAK
- 5-stage pipeline
 - Fetch, Decode, Execute, Memory Access, Writeback
 - Operand Forwarding - stall only on load-use hazard if enabled
 - Otherwise stall for all data hazards
 - Pipeline flushing for control hazards
- ELF file parsing with ‘elfio’ library
 - Compiles C programs to RISC-V ELF binary with riscv64-unknown-elf-gcc
- TUI interface
 - Displays disassembly with instruction pointer for each pipeline stage
 - Displays registers and stack, with highlighted changes since the last cycle

Planned Improvements

- Superscalar execution
 - Begin with 2-wide, later configurable
- Branch prediction
 - Support for different dynamic algorithms
- Reservation stations, reorder buffer & register renaming
 - Following Tomasulo's algorithm
- Support for 64-bit instructions (RV64)
- Support for floating point instructions ("F" extension)
- Support for vector instructions ("V" extension)

Planned Experiments & Benchmarks

- Performance of branch prediction algorithms
 - Compare dynamic algorithms (e.g. one and two-bit saturation counters)
 - Compare best dynamic to static not-taken and static taken
- Performance uplift of vector operations
 - Compare performance across benchmarks with vector extension enabled and disabled
- Evaluation of pipeline width across benchmarks
- Benchmarks
 - Sorting algorithms
 - Livermore Loops
 - Recursive algorithms